

Problem 3.17

You have \$1000 that you want to deposit in a savings account. Bank A computes the amount value of your investment using the amount function

$$A_1(t) = 1 + 0.049t,$$

whereas Bank B uses the amount function

$$A_2(t) = (1.4)^{12t}.$$

Where should you put your money?

Solution

We compare the value of a \$1000 deposit under each bank.

For Bank A,

$$A_1(2) = 1000(1 + 0.049(2)).$$

Thus,

$$A_1(2) = 1000(1.098) = 1098.$$

For Bank B,

$$A_2(2) = 1000(1.4)^{24}.$$

Thus,

$$A_2(2) \approx 3214199.7.$$

Therefore, you should put your money in Bank B, because in this bank the growth depends on time exponentially.